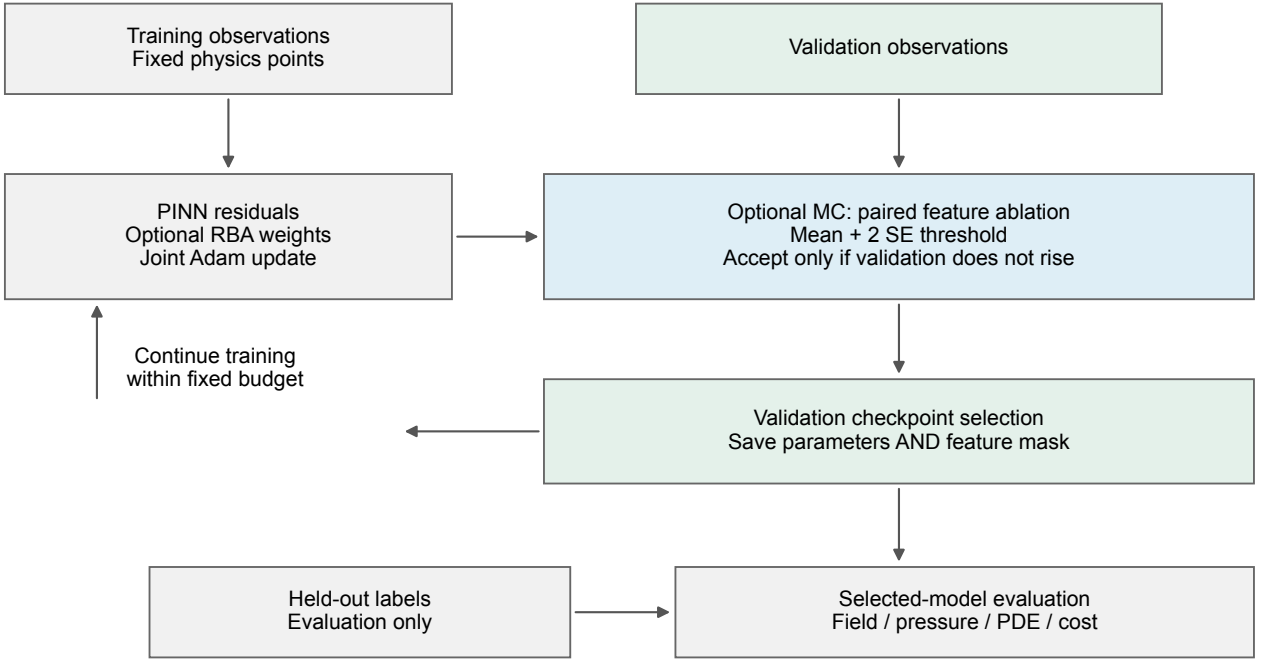
Training observations
Fixed physics points
Validation observations
PINN residuals
Optional RBA weights
Joint Adam update
Optional MC: paired feature ablation
Mean + 2 SE threshold
Accept only if validation does not rise
Continue training
within fixed budget
Validation checkpoint selection
Save parameters AND feature mask
Held-out labels
Evaluation only
Selected-model evaluation
Field / pressure / PDE / cost